



COLLABSPHERE

Enterprise Real-Time Collaborative Workspace & Project Intelligence Platform

PROJECT OVERVIEW

Project Title: COLLABSPHERE – Enterprise Real-Time Collaborative Workspace & Project Intelligence Platform

STATEMENT

Modern organizations rely on multiple disconnected applications for project management, document collaboration, communication, file storage, notifications, and team coordination. This fragmented workflow increases operational complexity, causes information silos, reduces productivity, and makes collaboration difficult across distributed teams.

The objective of **COLLABSPHERE** is to build an enterprise-grade collaborative workspace that unifies project planning, communication, documentation, file management, and real-time collaboration into a single platform.

Unlike traditional project management systems, **COLLABSPHERE** focuses on:

- Real-time collaboration
- Enterprise security
- Workspace isolation
- Permission-based access
- Version-controlled documents
- Event-driven architecture
- High scalability
- Modern user experience

The platform will support multiple organizations, multiple workspaces, simultaneous collaboration, live synchronization, intelligent notifications, and secure authentication.

OBJECTIVES

The project aims to:

- Build an enterprise-level collaboration platform.
- Eliminate disconnected project management tools.

- Provide Google Docs–style collaborative editing.
- Enable real-time Kanban synchronization.
- Support secure file sharing with version history.
- Provide integrated workspace communication.
- Offer enterprise-grade authentication and authorization.
- Ensure scalability for thousands of concurrent users.
- Maintain complete audit logs for all critical operations.

PRIMARY USE CASE

Organizations can create multiple workspaces, invite team members, assign roles, manage projects, collaborate on documents, communicate instantly, and track progress through synchronized Kanban boards.

Every workspace functions as an isolated collaborative environment where all project activities remain centralized.

UNIQUE FEATURES

Unlike a standard collaborative workspace, COLLABSPHERE includes several advanced enterprise capabilities.

1. REDIRECT-BASED GOOGLE AUTHENTICATION

Instead of using insecure popup authentication, the platform implements OAuth 2.0 Authorization Code Flow with PKCE.

Features:

- Google Redirect Login
- Secure OAuth callback
- Refresh Token support
- Session persistence
- Account linking
- Automatic profile synchronization
- JWT Access Tokens
- Secure Logout

2. ENTERPRISE AUTHENTICATION

Authentication methods include:

- Email & Password
- Google OAuth Redirect Login
- JWT Authentication
- Refresh Tokens
- Secure HTTP-only Cookies
- Password Reset
- Email Verification
- Session Expiration
- Device Session Management

3. MULTI-TENANT WORKSPACE ARCHITECTURE

Each organization operates independently.

Features:

- Organization isolation
- Workspace isolation
- Team separation
- Independent permissions
- Data segregation
- Secure tenant validation

4. ADVANCED ROLE-BASED ACCESS CONTROL

Roles:

- Super Admin
- Organization Admin
- Workspace Admin

- Project Manager
- Member
- Guest (Read Only)

Permissions include:

- Manage users
- Manage projects
- Invite members
- Delete workspace
- Edit documents
- Upload files
- View analytics
- Configure settings

5. SMART WORKSPACE DASHBOARD

Dashboard widgets include:

- Active Projects
- Recent Documents
- Pending Tasks
- Due Today
- Upcoming Deadlines
- Team Activity
- Notifications
- Workspace Statistics

6. DYNAMIC KANBAN BOARDS

Features:

- Drag-and-drop
- Multiple Boards

- Multiple Columns
- Swim lanes
- Task Labels
- Task Priorities
- Due Dates
- Checklist
- Attachments
- Assignees
- Time Tracking
- Activity History
- Real-time synchronization

7. REAL-TIME COLLABORATIVE DOCUMENT EDITOR

Google Docs–style collaboration.

Features:

- Simultaneous editing
- Cursor presence
- Live typing indicators
- Conflict resolution
- Operational Transform / CRDT synchronization
- Rich text formatting
- Tables
- Images
- Mentions
- Comments
- Suggestions
- Version history
- Document restore

8. INTEGRATED TEAM CHAT

Workspace includes:

- Global Chat
- Project Channels
- Direct Messages
- Group Conversations
- Typing Indicators
- Read Receipts
- File Sharing
- Emoji Reactions
- Mentions
- Thread Replies
- Message Search

9. TASK DISCUSSION SYSTEM

Each task contains:

- Comments
- Replies
- Attachments
- Mentions
- Activity Timeline
- Status Changes
- Audit History

10. FILE MANAGEMENT WITH VERSION CONTROL

Features:

- Folder hierarchy

- Drag-and-drop upload
- File previews
- Version history
- Restore previous versions
- File locking
- Download tracking
- Storage quota
- Permission control

11. SMART NOTIFICATION CENTER

Notifications include:

- Task Assigned
- Task Updated
- Mention
- Document Edited
- File Uploaded
- Chat Message
- Workspace Invitation
- Due Date Reminder

Delivery:

- Real-time Socket.IO
- Email Notifications
- In-App Notifications

12. ACTIVITY TIMELINE

Track every important event:

- User Login
- Task Updates

- Document Changes
- File Uploads
- Workspace Changes
- Role Changes
- Member Invitations

13. SEARCH SYSTEM

Global search across:

- Tasks
- Users
- Documents
- Chats
- Files
- Comments
- Workspaces

Supports:

- Filters
- Date Range
- User Search
- Full-text Search



14. DASHBOARD ANALYTICS

Analytics include:

- Task Completion Rate
- Productivity Trends
- Team Performance
- Active Users
- Document Activity

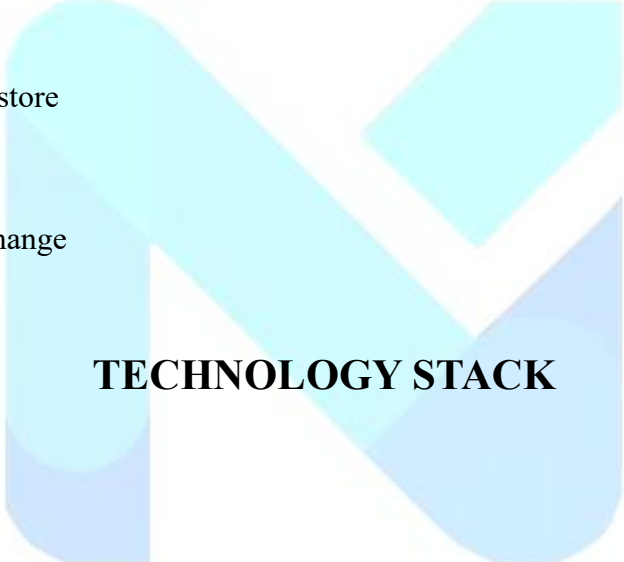
- Storage Usage
- Chat Statistics
- Workspace Growth

15. AUDIT LOGGING

Every important action is stored.

Examples:

- Login
- Logout
- File Upload
- Task Delete
- Document Restore
- Role Update
- Permission Change



TECHNOLOGY STACK

FRONTEND

- React.js
- TypeScript
- Vite
- Redux Toolkit
- React Query
- React Router
- Tailwind CSS
- Shadcn/Ui
- React Hook Form
- Zod
- Socket.IO Client

- TipTap Rich Text Editor
- React DnD
- Framer Motion

BACKEND

- Node.js
- Express.js
- TypeScript
- Socket.IO
- Prisma ORM
- PostgreSQL
- Redis
- BullMQ
- Passport.js
- Google OAuth 2.0
- JWT
- Multer
- Sharp
- Nodemailer



DEVOPS

- Docker
- Docker Compose
- GitHub Actions
- Nginx
- PM2
- AWS S3 Compatible Storage

SECURITY FEATURES

- OAuth 2.0 Authorization Code Flow
- PKCE Authentication

- JWT Authentication
- Refresh Tokens
- HTTP-only Cookies
- CSRF Protection
- Helmet Security
- CORS Protection
- Rate Limiting
- Input Validation
- SQL Injection Prevention
- XSS Prevention
- Password Hashing (Argon2/Bcrypt)
- Audit Logs
- File Type Validation

DATABASE MODULES

- Users
- Organizations
- Teams
- Workspaces
- Projects
- Boards
- Columns
- Tasks
- Documents
- Document Versions
- Chats
- Messages
- Files
- File Versions

- Notifications
- Audit Logs
- Sessions

WEEK-WISE DEVELOPMENT PLAN

WEEK 1 – FOUNDATION & AUTHENTICATION

BACKEND

- Project architecture setup
- PostgreSQL database schema
- Prisma ORM configuration
- JWT authentication
- Refresh Token implementation
- Redirect-Based Google OAuth Login
- Email verification
- Password reset
- Organization APIs
- Workspace CRUD APIs
- RBAC implementation

FRONTEND

- React project initialization
- Authentication pages
- Redirect Google Login flow
- Dashboard layout
- Workspace creation
- User profile
- Responsive navigation

WEEK 2 – REAL-TIME COLLABORATION

BACKEND

- Socket.IO setup
- Live Kanban synchronization
- Real-time document collaboration
- Presence management
- Typing indicators
- Conflict resolution
- Workspace synchronization

FRONTEND

- Drag-and-drop Kanban
- Live board updates
- Rich text editor
- Live cursor tracking
- User presence indicators
- Workspace switching

MID PROJECT REVIEW

BACKEND DELIVERABLES

- Authentication complete
- Google Redirect Login complete
- Workspace APIs complete
- Kanban APIs complete
- Real-time synchronization functional

FRONTEND DELIVERABLES

- Responsive dashboard
- Workspace management
- Live Kanban board
- Collaborative document editing

WEEK 3 – COMMUNICATION & FILE COLLABORATION

BACKEND

- Chat APIs
- Task comments
- File uploads
- Version control
- File permissions
- Search APIs
- Activity logging

FRONTEND

- Chat interface
- Threaded conversations
- Task comments
- File explorer
- Version history
- Search UI



WEEK 4 – ENTERPRISE FEATURES

BACKEND

- Notification service
- Email notifications
- Analytics APIs
- Audit logging
- Security hardening
- API optimization
- Final testing

FRONTEND

- Notification center
- Analytics dashboard
- Workspace settings
- Accessibility improvements
- Performance optimization
- Mobile responsiveness
- Final UI polishing

FINAL PROJECT REVIEW

BACKEND

- Enterprise-grade REST APIs
- Redirect Google OAuth Authentication
- JWT Authentication
- Real-time collaboration
- Secure RBAC
- Version-controlled files
- Notifications
- Audit logs
- Search engine
- Production-ready architecture

FRONTEND

- Fully responsive application
- Modern enterprise UI
- Real-time collaboration
- Live Kanban boards
- Google Docs–style editor
- Chat system
- Notification center

- Analytics dashboard
- File management
- Excellent user experience

EXPECTED OUTCOMES

Upon completion, COLLABSPHERE will function as a complete enterprise collaboration platform capable of supporting modern organizations with secure authentication, real-time teamwork, centralized project management, collaborative documentation, integrated communication, version-controlled file sharing, intelligent notifications, and scalable architecture. The solution demonstrates advanced full-stack engineering principles, distributed real-time systems, secure authentication using Google OAuth Redirect Flow, role-based access control, and production-ready software design suitable for enterprise environments.

**THE PROJECT REVIEW WILL BE CONDUCTED ON 03-09-2026,
AND PARTICIPANTS MUST SHARE THEIR SCREEN TO
DEMONSTRATE THE PROJECT.**